Multi-Level Analysis of Transcriptomic and Network Alterations in Papillary Renal Cell Carcinoma

FEDERICO LATTANZIO, Sapienza University of Rome, Italy

DOMENICO AZZARITO, Sapienza University of Rome, Italy

MICHELE PEZZA, Sapienza University of Rome, Italy

We analyzed TCGA-KIRP RNA-Seq data to study transcriptomic alterations and patient heterogeneity in papillary renal cell carcinoma. Differential expression analysis between paired tumor and normal samples identified genes with strong and significant expression changes. Network-based analyses revealed extensive disruption of gene co-expression patterns in tumors compared with normal tissue.

Patient Similarity Networks constructed from tumor expression profiles identified distinct molecular patient groups. Integration of transcriptomic and mutational data using Similarity Network Fusion further refined patient stratification and revealed communities associated with differences in survival, tumor mutational burden, and pathway alterations.

These results show that KIRP heterogeneity is reflected in coordinated transcriptomic and genomic changes that are effectively captured by network-based approaches.

Additional Key Words and Phrases: TCGA-KIRP, Differentially Expressed Genes (DEGs), Co-Expression Networks, Patient Similarity Networks

1 Introduction

Renal carcinoma accounts for about 3% of adult cancers and includes several subtypes with different biological and clinical features. Among them, Kidney Renal Papillary Cell Carcinoma (KIRP) is the second most common subtype and is characterized by specific molecular changes affecting gene expression and regulation. Comparing gene-expression profiles between KIRP tumors and matched normal tissues is an important approach for understanding tumor development and for identifying markers that may have diagnostic, prognostic, or therapeutic value.

Differential expression analysis is usually the first step in transcriptomic studies, as it highlights genes whose expression changes between conditions. However, this approach alone does not fully describe cancer biology, because genes do not act independently but interact within complex regulatory networks. For this reason, it is necessary to study not only changes in single-gene expression levels but also how relationships between genes are altered in the tumor state.

Furthermore, gene co-expression and differential co-expression network analyses make it possible to investigate these interaction changes at a system level. Such analyses can reveal the loss of normal network organization and the appearance of tumor-specific regulatory hubs. In addition, transcriptomic data can be used to compare patients by building Patient Similarity Networks, which help identify molecular subgroups and explore intra-tumor heterogeneity [5].

Previous studies have reported that KIRP shows widespread alterations in metabolic processes, signaling pathways, and epigenetic regulation. This suggests that tumor progression is driven by coordinated changes in gene interactions rather than by isolated expression shifts. Based on these observations, we hypothesize that papillary renal carcinoma involves a global reorganization of gene co-expression patterns and that these changes are reflected in the molecular similarity between patients.

The aim of this study is to characterize transcriptomic alterations in KIRP at multiple levels by integrating differential expression analysis with co-expression networks, differential co-expression networks, and Patient Similarity Networks. This integrated approach is intended to provide biologically meaningful insights that may be relevant for future translational and clinical research.

2 Materials and Methods

Data Preprocessing

Gene-expression data were obtained from the TCGA-KIRP project through the Genomic Data Commons (GDC) portal. Samples corresponding to the Primary Tumor and Solid Tissue Normal categories of the STAR–Counts workflow within the Transcriptome Profiling section were selected. Data download and preparation were performed using the TCGAbiolinks package, which allowed the formulation of queries for the two sample types, the download of standardized files, and the conversion of each dataset into a SummarizedExperiment object containing the raw count matrix and gene annotation, including genomic coordinates, Ensembl identifiers, gene names, and functional categories. Consistency of gene annotation between tumor and normal samples was verified by comparing the rowRanges and confirming that the same genes were represented in both conditions.

To ensure a biologically meaningful comparison, TCGA barcodes were truncated to the first twelve characters, which uniquely identify each patient. Patients were included only when a single Primary Tumor sample and at least one corresponding Solid Tissue Normal sample were available. This selection yielded a cohort of thirty-two paired patients, allowing for an accurate evaluation of intra-individual transcriptomic differences.

Authors' Contact Information:, lattanzio.1886519@studenti.uniroma1.it; Federico Lattanzio, Sapienza University of Rome, Italy, Rome, azzarito.2209000@studenti.uniroma1.it; Domenico Azzarito, Sapienza University of Rome, Italy, Rome, pezza.1845897@studenti.uniroma1.it; Michele Pezza, Sapienza University of Rome, Italy, Rome.

Before normalization, the raw count matrix underwent two filtering procedures. The first step consisted of removing genes with zero counts in any tumor or normal sample, preventing instability in fold-change estimation and retaining only genes expressed reliably across all samples. The second step consisted of applying an abundance filter in which genes were retained only if they had at least ten counts in at least ninety percent of samples, thereby excluding genes dominated by technical noise rather than biological signal.

Tumor and normal samples were merged into a single matrix and used to construct a DESeqDataSet object. Normalization was performed by estimating the DESeq2 size factors, which correct for differences in sequencing depth and global shifts in count distributions. The normalized count matrix was then separated again into tumor and normal subsets while preserving the ordering of genes and patients. Additional checks confirmed the absence of missing or non-numeric values, the alignment of gene rows across the two conditions, and the consistency of sample names in all matrices.

For gene–gene and patient–patient correlation analyses, normalized matrices were transformed using the function log2(x + 1) to stabilize variance and limit the influence of outliers. The final set of genes used in differential expression analysis, co-expression network construction, and the development of Patient Similarity Networks was obtained by applying the filtering, normalization, and quality-control steps described above.

Differentially Expression Analysis

Differential expression analysis was performed using the normalized count matrices obtained after preprocessing. The objective was to identify genes whose transcriptional activity differs systematically between tumor tissues and paired normal tissues belonging to the same patients.

For each gene, the $\log_2$ fold change ($\log_2$FC) was computed as the ratio between the mean expression levels in the two conditions:

$$\log_2\mathrm{FC} = \log_2\left(\frac{\overline{X}_{\text{tumor}}}{\overline{X}_{\text{normal}}}\right),$$

where $\overline{X}_{\text{tumor}}$ and $\overline{X}_{\text{normal}}$ denote the mean normalized values for each gene in tumor and normal samples. This metric provides a directional quantification of transcriptomic variation and represents the main parameter for the biological interpretation of differential changes.

To evaluate whether the observed differences were statistically significant, a paired t-test was applied independently to each gene. The paired design controls for inter-individual variability by exploiting the correspondence between tumor and normal samples from the same patient. The hypotheses considered were:

$$H_0 : \mu_{\text{tumor}} = \mu_{\text{normal}}, \qquad H_1 : \mu_{\text{tumor}} \neq \mu_{\text{normal}}.$$

P-values were adjusted for multiple testing using the Benjamini–Hochberg procedure, which controls the false discovery rate.

The set of differentially expressed genes was defined using two simultaneous criteria. The first criterion was statistical significance, expressed as an adjusted p-value satisfying

$$\mathrm{FDR} \leq 0.05.$$

The second criterion was the magnitude of biological effect, requiring

$$|\log_2\mathrm{FC}| \geq 2.$$

These conditions ensure that selected genes exhibit both a substantial change in expression and robust statistical evidence.

The list of differentially expressed genes obtained through this procedure served as the basis for subsequent analyses.

Co-Expression Network Construction

Gene co-expression networks were constructed separately for cancer and normal samples using the 724 differentially expressed genes (DEGs) identified in the previous step. Prior to correlation analysis, normalized expression values were log2-transformed to stabilize variance and improve the normality of the data distribution.

Pairwise Pearson correlation coefficients were computed between all gene pairs across samples using the `corr.test` function from the `psych` R package, which simultaneously provides p-values with false discovery rate (FDR) adjustment using the Benjamini-Hochberg procedure. This approach is consistent with established co-expression network methodologies that rely on Pearson correlation as the primary similarity measure.

Binary adjacency matrices were constructed by applying dual thresholds: (1) a correlation coefficient threshold ($|\rho| \geq 0.7$ for cancer, $|\rho| \geq 0.8$ for normal) to ensure biologically meaningful associations, and (2) a statistical significance threshold (FDR-adjusted $p \leq 0.05$) to control false positive rates. An edge was retained in the network only if both conditions were satisfied. Different correlation thresholds were applied to achieve comparable network densities between conditions while maintaining stringent statistical criteria.

Scale-Free Topology Assessment

Biological networks typically exhibit scale-free properties, characterized by a power-law degree distribution $P(k) \sim k^{-\gamma}$, where most nodes have few connections while a small number of hub nodes are highly connected. To verify this property, we plotted log-log degree distributions and estimated the power-law exponent γ using Maximum Likelihood Estimation (MLE) with a truncated zeta function as the normalization constant.

Hub Gene Identification

Hub genes were defined as nodes with degree values in the top 5% of all connected nodes within each network. This percentile-based approach is widely used in co-expression network analysis to identify genes with central regulatory roles. Additionally, betweenness centrality was computed using the `igraph` R package to identify

genes that serve as bridges between different network modules. Genes with high betweenness centrality control information flow through the network and may represent critical regulatory nodes.

Differential Co-Expression Network Construction

The differential co-expression network was constructed to identify gene pairs whose correlation patterns changed significantly between normal and cancer conditions, independent of changes in expression levels. This approach reveals regulatory rewiring events that may drive cancer phenotypes.

Fisher's Z-transform was applied to convert Pearson correlation coefficients to approximately normally distributed Z-scores:

$$z = \frac{1}{2} \ln \left(\frac{1+r}{1-r} \right) \tag{1}$$

The difference in correlations between conditions was tested using the Z-statistic:

$$Z_{diff} = \frac{z_{cancer} - z_{normal}}{\sqrt{\frac{1}{n_c - 3} + \frac{1}{n_n - 3}}} \tag{2}$$

where n_c and n_n are the sample sizes for cancer and normal conditions, respectively. A stringent threshold of $|Z_{diff}| \geq 5$ was applied to identify gene pairs with statistically significant changes in co-expression.

Signed Differential Network Analysis

To distinguish between edges where correlations increased versus decreased in cancer, the differential network was decomposed into two signed subnetworks:

- **Positive subnetwork**: Gene pairs with $Z_{diff} \geq 5$ (correlations strengthened in cancer)
- **Negative subnetwork**: Gene pairs with $Z_{diff} \leq -5$ (correlations weakened in cancer)

This signed analysis enables identification of genes undergoing distinct types of regulatory rewiring during tumorigenesis.

Patient Similarity Network

To identify subgroups of patients characterized by distinct molecular profiles, a Patient Similarity Network (PSN) approach was adopted. This methodology models the cohort as a graph where each node represents a patient and edges connect individuals showing high biological resemblance. The networks were constructed using the gene expression profiles of the previously identified Differentially Expressed Genes (DEGs), a choice designed to focus the topology on transcriptomic signals strictly relevant to the disease pathology rather than global expression noise. Prior to network construction, RNA-Seq counts were normalized using Counts Per Million (CPM) and transformed using $\log_2(x+1)$ to ensure comparability.

Network Construction and Multi-omics Integration

The topology of the networks was defined by quantifying the pairwise similarity between patients using Pearson correlation. To retain only cooperative biological signals and facilitate community detection, self-loops were removed and negative correlations were excluded from the adjacency matrices.

This framework was applied to construct two baseline networks: a **Cancer PSN**, built using primary tumor samples to capture disease heterogeneity, and a **Normal PSN**, derived from solid tissue normal samples. The latter served as a biological reference to evaluate the stability and specificity of the clusters identified in the tumor cohort.

To further refine patient stratification by integrating genomic heterogeneity, **Similarity Network Fusion (SNF)** was applied to the cancer cohort. This advanced technique fused the gene expression-based similarity matrix with a second layer derived from the patients' somatic mutational profiles. The resulting fused network represents a global consensus of patient similarity, capturing both downstream transcriptomic patterns and upstream genomic alterations, thereby providing a more comprehensive view of the tumor landscape.

Community Detection and Characterization

Subgroups of patients were identified by applying the **Louvain community detection algorithm** to the generated networks (Cancer, Normal, and SNF). This method optimizes network modularity to partition patients into distinct communities without the need for arbitrary thresholds or a pre-defined number of clusters.

Following the identification of these communities, a rigorous characterization was performed to assess clinical and biological relevance of communities obtained from tumor gene expression data. The analysis was restricted to the full cohort of 290 tumor samples in order to ensure adequate statistical power, as such analyses would not have been feasible when limited to paired tumor–normal samples. The molecular subtypes were integrated with clinical metadata to evaluate potential differences in prognosis (Overall Survival) using Kaplan-Meier analysis and the log-rank test. Additionally, we assessed the association between communities and key clinical variables, including demographic characteristics and Tumor Mutational Burden (TMB).

Statistical significance was evaluated using the Kruskal–Wallis test for continuous variables and Fisher's exact test for categorical variables, applying the Benjamini–Hochberg correction for multiple comparisons. Finally, to elucidate the biological drivers specific to each subgroup, differential expression analysis was conducted for each community against the rest of the cohort using `limma`. The resulting gene signatures were then mapped to biological processes and pathways through Gene Ontology (GO) and KEGG enrichment analyses, linking the topological communities to specific functional phenotypes.

Gene Set Enrichment Analysis

To provide a biological characterization of the key regulatory elements identified in the co-expression analysis, a Gene Set Enrichment Analysis (GSEA) was performed on the set of cancer-specific hubs. This analysis aimed to determine whether the highly connected genes in the tumor network were significantly associated with specific biological processes or signaling pathways.

The functional enrichment was conducted using the clusterProfiler R package. The compareCluster function was used to perform parallel enrichment analysis on cancer and normal hubs, utilizing the enrichGO function for Gene Ontology (GO) annotations of Biological Processes and the enrichKEGG function for KEGG (Kyoto Encyclopedia of Genes and Genomes) pathway analysis.

A significance threshold of p-value $\leq$ 0.05 was applied to filter enriched terms and pathways. Gene symbols were converted to ENTREZ IDs for compatibility with the annotation sources. This analysis enabled the interpretation of network topology in the context of known biological functions and disease-associated mechanisms.

3 Results and Discussions

Data Acquisition and Cohort Composition

The analysis was conducted using RNA-Seq data from the TCGA-KIRP project, including paired primary tumor and solid normal tissue samples. After sample selection, harmonization, and quality-control procedures, the dataset used for differential expression analysis comprised 32 patients for whom both transcriptomic profiles were available. This design enabled direct intra-patient comparisons and reduced non-biological inter-individual variability.

The initial dataset included 60,660 genes, consistently annotated across the two conditions. Following the removal of genes with zero counts in at least one sample and the application of expression filters aimed at eliminating weakly informative signals, 15,886 genes robustly expressed in the majority of samples were retained. This preprocessing strategy allowed the analysis to focus on a subset of genes that are both statistically reliable and biologically meaningful, improving the stability of expression estimates and comparative statistical tests.

Overall, the resulting cohort exhibits a technically coherent and well-balanced structure, providing a solid foundation for the identification of transcriptomic alterations associated with papillary renal carcinoma.

Differential Expression Analysis

Differential expression analysis between paired tumor and normal tissues revealed a profound reorganization of the transcriptomic profile in papillary renal carcinoma. By applying a paired statistical test with multiple-testing correction using the False Discovery Rate (FDR) and imposing an effect-size threshold of $|\log_2 \mathrm{FC}| \geq 2$, a total of 724 differentially expressed genes (DEGs) were identified.

The joint distribution of statistical significance and biological effect size is summarized in the volcano plot (Figure X). Most genes cluster around $\log_2 \mathrm{FC} \approx 0$ and low values of $-\log_{10}(\mathrm{FDR})$, indicating that a large fraction of the transcriptome does not undergo substantial changes between the two conditions. Nevertheless, a distinct subset of genes simultaneously exceeds both the significance and effect-size thresholds, highlighting robust and biologically relevant transcriptional alterations.

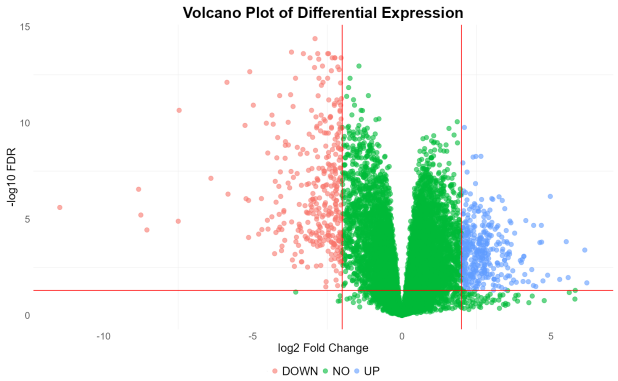

Fig. 1. Volcano Plot

The volcano plot in Figure 1 further reveals an asymmetric distribution of DEGs between upregulated and downregulated genes, with the presence of genes exhibiting large fold changes and high statistical significance in both directions. This pattern suggests that neoplastic transformation in KIRP is not associated with a global and uniform shift in gene expression, but rather with the selective differential regulation of specific transcriptional programs. The presence of highly significant genes displaying moderate expression changes underscores the importance of jointly considering statistical significance and biological effect size when interpreting differential expression results.

From a biological perspective, the identification of a substantial set of DEGs indicates that papillary renal carcinoma is characterized by coordinated remodeling of gene activity rather than by isolated alterations of a few key genes. This observation is consistent with the hypothesis that tumorigenesis involves complex processes, including metabolic reprogramming, loss of normal renal tissue functions, and activation of adaptive cellular responses.

Although differential expression analysis provides a fundamental description of changes in mean gene-expression levels, it does not directly capture alterations in functional relationships among genes. Nevertheless, the identified set of DEGs constitutes a crucial informative panel that serves as the basis for subsequent network analyses aimed at investigating how neoplastic transformation in KIRP is reflected in a systemic reorganization of transcriptional interactions.

Co-Expression Network Topology

Co-expression networks were successfully constructed for both cancer and normal conditions. The cancer network contained 4,724 edges among 439 connected nodes (60.6% of DEGs), while the normal network contained 8,243 edges among 488 connected nodes (67.4% of DEGs). The higher edge count in the normal network despite a more stringent correlation threshold ($|\rho| \geq 0.8$ vs. $|\rho| \geq 0.7$) suggests that normal tissue maintains tighter transcriptional coordination compared to tumor tissue, consistent with previous reports of regulatory disruption in cancer [13].

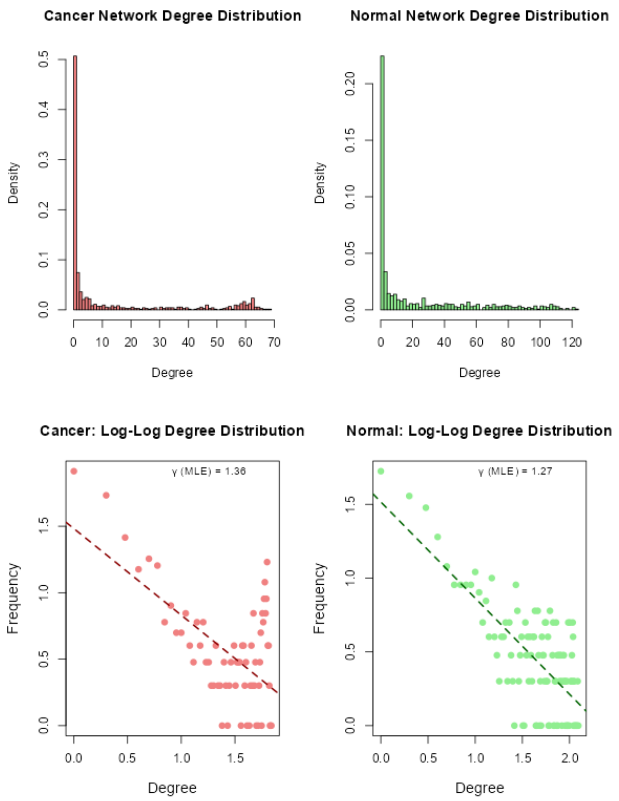

Fig. 2. Cancer and Normal network degree distribution

Both networks exhibited scale-free properties, as evidenced by their power-law degree distributions, which confirms that they contain a small number of highly connected hub genes and many poorly connected peripheral genes. The estimated power-law exponents (γ) were 1.36 for the cancer network and 1.27 for the normal network. Although these values fall below the typical range for biological scale-free networks ($2 < \gamma < 3$), the heavy right tail of the distributions supports the presence of a hub-based topology. The deviation is likely attributable to the relatively small sample size available for this specific tumor subtype; nonetheless, the identified topology remains consistent with a network architecture that confers robustness against random perturbations while creating vulnerability to targeted attacks on hub nodes.

Network density was notably different between conditions (cancer: 0.0180; normal: 0.0315), indicating that tumors exhibit reduced global connectivity among DEGs. This observation aligns with the hypothesis that cancer disrupts normal regulatory relationships, leading to a loss of coordinated gene expression patterns [11].

Hub Gene Identification and Comparison

Hub genes were identified as nodes in the top 5% by degree in each network. The cancer network contained 30 hubs (threshold: degree $\geq$ 63), while the normal network contained 27 hubs (threshold: degree $\geq$ 106). Strikingly, no genes maintained hub status in both conditions, indicating complete rewiring of central regulatory nodes between normal and tumor states.

The top three hub genes in the **cancer network** were:

(1) **LAIR1** (Leukocyte-associated immunoglobulin-like receptor 1) - Degree: 69

(2) **MYO1F** (Myosin IF) - Degree: 68

(3) **LAPTM5** (Lysosomal protein transmembrane 5) - Degree: 67

These genes are predominantly associated with immune cell function and signaling. LAIR1 is an inhibitory receptor expressed on immune cells that regulates inflammatory responses. The emergence of immune-related hubs in the cancer network is consistent with the well-documented role of immune infiltration and tumor-immune interactions in kidney cancer [12].

The top three hub genes in the **normal network** were:

(1) **SLC34A1** (Solute carrier family 34 member 1) - Degree: 124

(2) **SLC13A3** (Solute carrier family 13 member 3) - Degree: 122

(3) **MME** (Membrane metalloendopeptidase/Neprilysin) - Degree: 122

These genes are characteristic of normal kidney tubular epithelial function. SLC34A1 and SLC13A3 are sodium-dependent phosphate and dicarboxylate transporters essential for renal reabsorption, while MME is a kidney brush border enzyme. The loss of these metabolic and transport-related hubs in cancer reflects the dedifferentiation of tumor cells away from normal kidney epithelial phenotypes [9].

Centrality Measure Comparison

Comparison of degree centrality with betweenness centrality revealed moderate overlap between the two measures. In the cancer network, Spearman correlation between degree and betweenness was 0.812, while in the normal network it was 0.850. However, only 6.8% and 2.2% of hub genes (by either measure) overlapped in cancer and normal networks, respectively, as we can see from the Figure 3.

Genes that are hubs by both degree and betweenness criteria represent particularly critical regulatory nodes, as they are both highly connected and serve as bridges between different network modules. Conversely, genes with high betweenness but low degree may connect disparate functional modules and represent potential therapeutic targets whose disruption could fragment oncogenic signaling cascades [15].

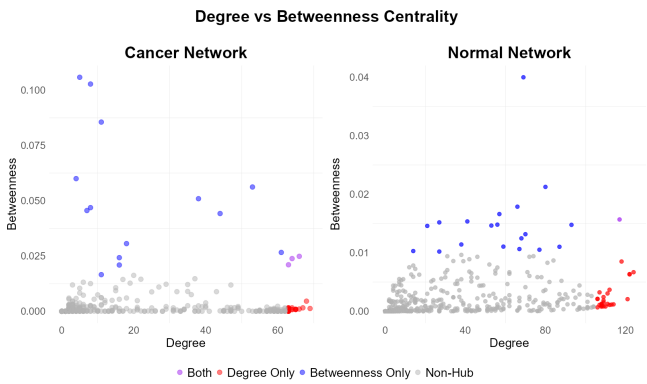

Fig. 3. Degree vs Betweenness Centrality

Differential Co-Expression Network Analysis

The differential co-expression network contained 3,894 edges representing gene pairs with significantly altered correlations between normal and cancer conditions ($|Z| \geq 5$). This network captured regulatory rewiring events independent of expression level changes [1], providing complementary information to traditional differential expression analysis [6].

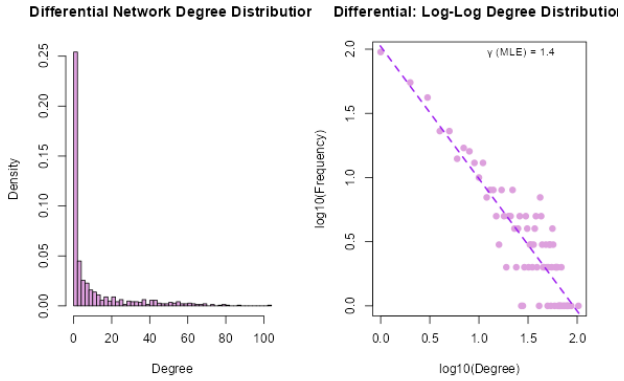

Fig. 4. Differential Co-Expression Network Degree Distribution

Analysis of differential degree revealed 28 differential hubs, with 14 overlapping with normal network hubs and none overlapping with cancer network hubs. This asymmetry indicates that genes serving as hubs in normal tissue underwent substantial rewiring in cancer, often losing their coordinating role.

The top three differential hubs were:

(1) **STRA6** (Stimulated by retinoic acid 6) - Differential Degree: 103

(2) **PODXL** (Podocalyxin-like) - Differential Degree: 87

(3) **CALB1** (Calbindin 1) - Differential Degree: 81

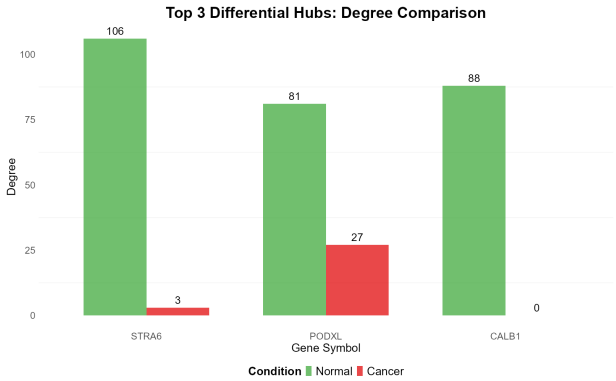

Fig. 5. How connectivity changes from Normal to Cancer.

STRA6: A Case Study in Regulatory Rewiring

STRA6, the top differential hub, exhibited dramatic connectivity changes: from 106 connections in normal tissue to only 3 in cancer, with 103 edges lost and zero preserved. STRA6 encodes a membrane receptor for retinol-binding protein and plays a critical role in vitamin A uptake and signaling [7]. The complete loss of STRA6's hub status in cancer suggests disruption of retinoid metabolism, which is known to regulate cellular differentiation and proliferation. Similar dysregulation of STRA6 has been reported in papillary thyroid carcinoma, where it regulates the tumor immune microenvironment [14], and in colorectal cancer, where the RBP4-STRA6 axis maintains cancer stem cells [2].

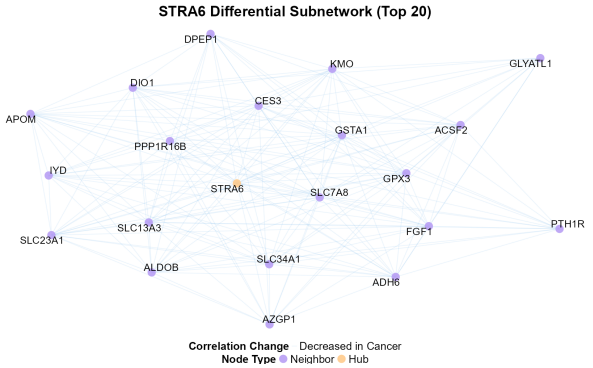

Fig. 6. STRA6 differential subnetwork, displaying the top 20 neighbors of STRA6 in the differential network.

Visual inspection of the STRA6 differential subnetwork (Figure 6) reveals that while STRA6 is isolated from other broad differential hubs, it remains tightly linked to specific *negative* hubs such as GPX3 and CES3. This structural isolation indicates that STRA6 does not merely lose random connections; rather, it is part of a specific functional module, likely related to kidney metabolic homeostasis, that collectively disintegrates during tumorigenesis. The presence of neighbors like GPX3 (an antioxidant enzyme) and CES3 suggests a coordinated loss of metabolic and protective functions shared

across this specific gene cluster, distinguishing it from other rewiring events in the network.

PODXL and CALB1: Additional Rewired Regulators

PODXL (degree change: 81 → 27) is a transmembrane sialomucin associated with poor prognosis in multiple cancers, including colorectal and renal cell carcinoma [10]. Its partial loss of connectivity may reflect altered adhesion and signaling properties during tumor progression.

CALB1 exhibited complete connectivity loss (88 → 0), consistent with its role as a calcium-binding protein involved in normal epithelial cell function. Loss of CALB1 co-expression relationships may indicate dedifferentiation of tumor cells [8].

Signed Differential Network Analysis

Decomposition of the differential network revealed an asymmetry: 2,905 edges showed decreased correlation in cancer (74.6%), while only 989 edges showed increased correlation (25.4%). This predominance of lost correlations supports the model of cancer as a state of regulatory breakdown rather than acquisition of new coordinated programs [13].

The top positive hubs (correlations increased in cancer) included:

- **LPL** (Lipoprotein lipase) - Positive Degree: 53
- **SLFN13** (Schlafen family member 13) - Positive Degree: 39
- **HSPA2** (Heat shock protein family A member 2) - Positive Degree: 36

The top negative hubs (correlations decreased in cancer) included:

- **STRA6** - Negative Degree: 81
- **GPX3** (Glutathione peroxidase 3) - Negative Degree: 76
- **CES3** (Carboxylesterase 3) - Negative Degree: 75

GPX3 is notable as an antioxidant enzyme frequently silenced in renal cell carcinoma, suggesting that loss of oxidative stress response coordination may contribute to tumorigenesis [3]. The emergence of heat shock proteins (HSPA2) among positive hubs is consistent with increased stress response activation in cancer.

Biological Implications

The complete lack of overlap between cancer and normal hub sets, combined with the predominance of lost correlations in the differential network, suggests that KIRP is characterized by profound disruption of normal transcriptional coordination. The transition from kidney-specific transporters (SLC34A1, SLC13A3, MME) to immune-related hubs (LAIR1, MYO1F, LAPTM5) reflects both loss of epithelial differentiation and increased immune infiltration characteristic of kidney cancers.

The identification of STRA6 as the top differential hub provides a potential mechanistic link to retinoid metabolism dysregulation, which has emerging significance in cancer biology. Retinoids regulate cellular differentiation, and STRA6 disruption may contribute to the dedifferentiated phenotype of KIRP tumors [7].

These findings demonstrate the power of differential co-expression analysis to reveal regulatory rewiring events invisible to traditional differential expression approaches, identifying novel candidate genes and pathways for therapeutic intervention in kidney renal papillary cell carcinoma.

Gene Set Enrichment Analysis

To elucidate the biological implications of the network rewiring, we performed a functional enrichment analysis comparing the hub genes identified in the normal and cancer networks. As shown in Figure 7, the Gene Ontology (GO) analysis highlights a distinct functional dichotomy. The normal tissue hubs are exclusively enriched in processes maintaining physiological renal functionality, such as the responsiveness to toxic substances and kidney development. In contrast, the cancer hubs show a predominant enrichment in processes related to immune response, leukocyte activation, and inflammation. This suggests that the tumor microenvironment in KIRP is characterized by a loss of specific renal functions and a strong infiltration of immune cells, establishing a chronic inflammatory state as we would expect.

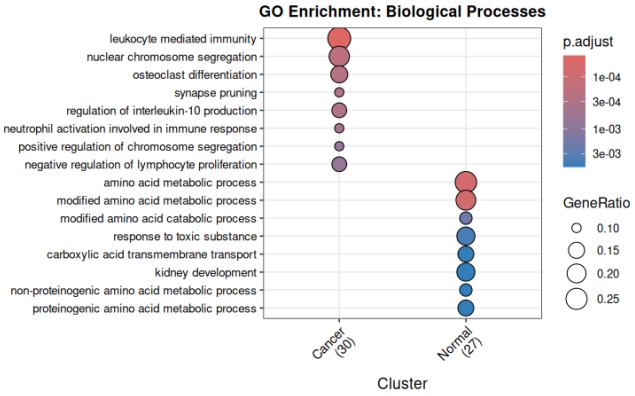

Fig. 7. GO Enrichment for biological processes between normal and cancer tissues

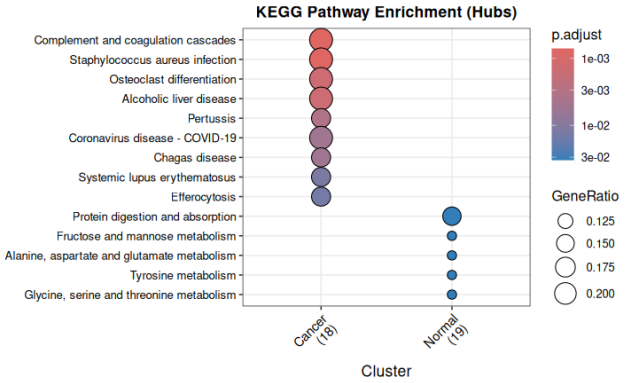

Fig. 8. KEGG Pathways Enrichment between normal and cancer tissues

This functional shift is further corroborated by the KEGG pathway analysis presented in Figure 8, where 18 cancer hubs and 19 normal hubs were successfully mapped. The pathways associated with normal hubs confirm the metabolic nature of healthy kidney tissue, focusing on the reabsorption of proteins and sugars to prevent urinary loss. Conversely, the cancer network reveals a radical shift towards immune modulation and tissue remodeling. Dominant pathways such as Osteoclast differentiation and Staphylococcus aureus infection indicate a highly inflamed microenvironment where hubs no longer support filtration but rather drive structural changes and recruit immune cells to sustain tumor progression.

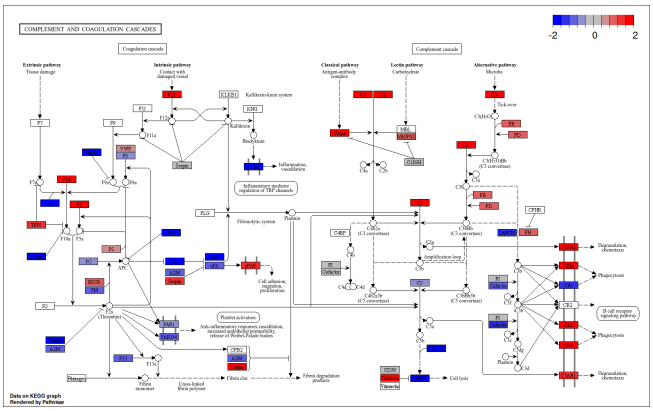

Fig. 9. Final Pathway of "Complement and coagulation cascades"

Finally, Figure 9 details the "Complement and coagulation cascades," a key pathway enriched in the cancer network. The map reveals the upregulation of critical components of the classical complement pathway, including the initiator C1q and the central hub C3 (red nodes). Crucially, the strong upregulation of receptors such as C5aR1, C3AR1, and integrin receptors (CR3/CR4) points towards active signaling for immune cell recruitment (chemotaxis) and adhesion, rather than direct cell lysis, as the terminal MAC complex remains downregulated, creating right conditions that sustain chronic inflammation and tumor progression. Furthermore, the upregulation of uPAR (PLAUR) and coagulation factors like F10 and F12 highlights a mechanism for extracellular matrix degradation and a pro-thrombotic state, potentially facilitating tumor invasiveness and metastasis.

Patient Similarity Network

Community Detection in Expression-based Networks

Using gene expression data from cancer samples alone, the analysis initially identified two distinct patient communities as shown in Figure 10. However, Figure 11 shows thst upon integrating mutational profiles via Similarity Network Fusion (SNF), the stratification resolution improved, revealing a total of three multi-omics communities. This suggests that the addition of genomic data captured finer biological substructures that were not resolvable by transcriptomic profiles alone.

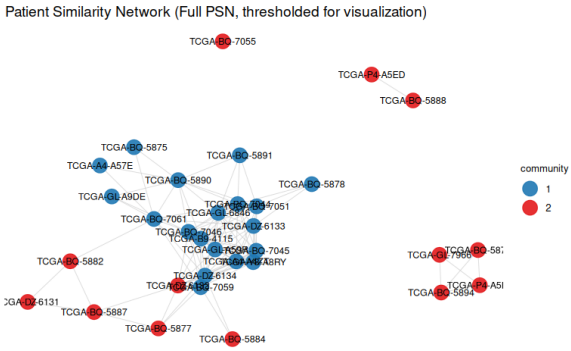

Fig. 10

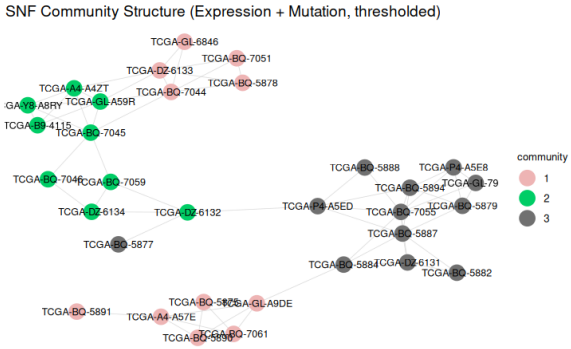

Fig. 11

Patient Similarity Network Based on Normal Gene Expression

The Patient Similarity Network built from normal renal gene expression profiles shows a simple structure, with two main communities identified by the Louvain algorithm. The network is highly cohesive, with strong connections within communities and clear separation between them, indicating that normal transcriptomic profiles are largely conserved across patients.

When compared with the tumor PSN using patients with both profiles available, a clear reorganization of similarity relationships is observed. Patients grouped together in the normal PSN are often assigned to different communities in the tumor PSN, showing that similarity patterns present in healthy tissue are not preserved in the tumor state.

This divergence suggests that tumor development introduces new molecular variability that overrides normal physiological differences. While the normal PSN reflects stable baseline expression programs, the tumor PSN captures the molecular heterogeneity of papillary renal carcinoma and provides a more informative framework for identifying biologically relevant patient subgroups.

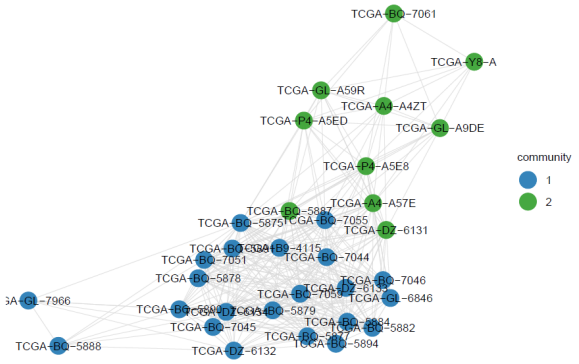

Fig. 12. PSN from Normal Samples: 0.85 threshold

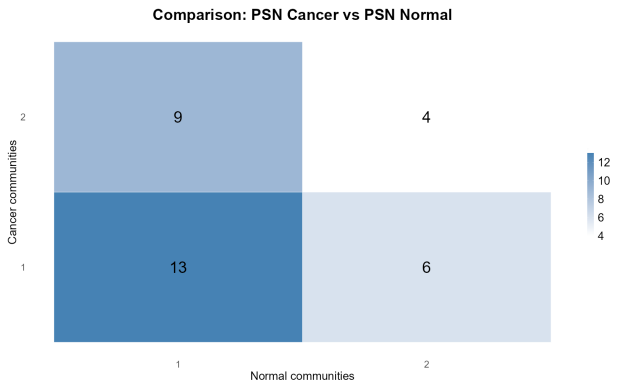

Fig. 13. Community Correspondence Between Cancer and Normal PSNs

Similarity Network Fusion

Exclusively utilizing gene expression data, the analysis initially partitioned the cohort into two broad communities. However, the integration of mutational profiles via Similarity Network Fusion (SNF) significantly enhanced the stratification resolution, identifying a total of three multi-omics communities.

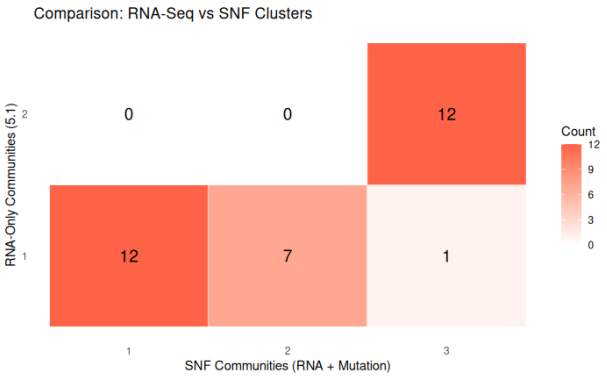

Fig. 14. Community Correspondence Between SNF and RNA only Communities

As illustrated by the confusion matrix in Figure 14, this transition was not uniform: while the transcriptomic 'Cluster 2' proved to be robust—mapping consistently to a single multi-omics group (SNF Cluster 3)—the larger 'Cluster 1' was dissected into two distinct subgroups (SNF Clusters 1 and 2). This finding suggests that identical transcriptomic phenotypes can mask underlying genomic heterogeneity, which is successfully deconvoluted only when somatic mutations are integrated into the network topology.

PSN Communities Characterization

The Patient Similarity Network (PSN) constructed from tumor gene expression profiles revealed a clear modular organization, identifying three main transcriptomic communities within the KIRP cohort. This structure indicates that tumor heterogeneity is driven by coordinated expression programs rather than isolated gene-level changes.

To further refine patient stratification, transcriptomic similarity was integrated with mutational profiles using Similarity Network Fusion (SNF). The fused network displayed increased resolution, partitioning patients into six distinct molecular communities. This result suggests that genomic alterations provide complementary information to expression data, enabling the identification of patient subgroups that are not fully distinguishable using transcriptomic profiles alone. The SNF communities therefore represent integrated molecular phenotypes reflecting both transcriptional and mutational variability.

Analysis and Clinical Enrichment

Survival analysis revealed significant differences in overall survival across SNF communities (Kaplan-Meier analysis, log-rank test $p = 0.003$), despite survival information not being used during clustering. This supports the biological and clinical relevance of the identified molecular stratification. Tumor mutational burden (TMB), with a median value of approximately 1.04 mutations per megabase, differed significantly among communities, indicating a role for genomic instability in shaping patient heterogeneity. Consistently, enrichment analysis identified significant associations between specific communities and alterations in DNA damage response (DDR) and RTK/RAS pathways, while no significant enrichment was observed for TP53, WNT, or PI3K pathways.

Functional enrichment analysis further highlighted marked biological differences among communities. Several groups were enriched for immune-related processes, including leukocyte migration, cytokine signaling, and antigen presentation, suggesting variable immune involvement and tumor–microenvironment interactions. Other communities were characterized by enrichment in metabolic and transport-related processes, such as organic acid metabolism, small molecule catabolism, and ion transport, consistent with metabolic reprogramming and altered renal-specific functions. KEGG pathway analysis supported these findings, revealing immune-related pathways in some communities and pathways related to cell adhesion, extracellular matrix interaction, and focal adhesion in others, pointing to differences in tumor structure and invasive potential.

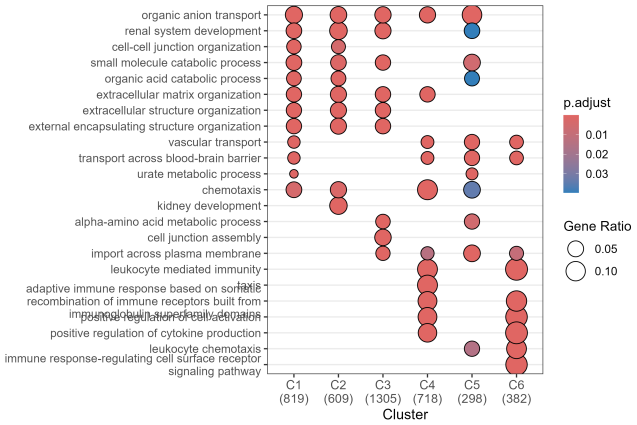

Fig. 15. GO Biological Process Enrichment Across SNF Communities

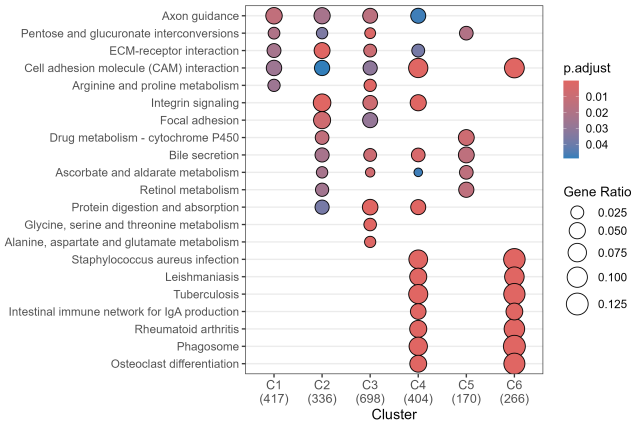

Fig. 16. KEGG Pathway Enrichment Across SNF Communities

Overall, these results demonstrate that PSN-based analysis, particularly when combined with mutational data through SNF, identifies biologically and clinically meaningful KIRP subtypes. The observed differences in transcriptional organization, genomic alterations, survival outcomes, and functional programs support a systems-level view of tumor heterogeneity and suggest that KIRP progression arises from coordinated alterations across multiple molecular layers rather than isolated gene-level changes.

4 Conclusions

In this study, we performed a multi-level analysis of papillary renal cell carcinoma using TCGA-KIRP data, combining differential expression, network-based approaches, and patient similarity modeling. Rather than focusing only on single-gene changes, we investigated how transcriptional relationships and patient-level similarities are reorganized in the tumor state.

Differential expression analysis identified a robust set of genes showing strong and significant changes between tumor and normal tissues. Network analyses revealed that these changes are accompanied by a marked disruption of gene co-expression structure in cancer, with reduced connectivity and a complete reorganization of hub genes. Normal kidney networks were dominated by transport and metabolic genes, while tumor networks showed the emergence of immune-related hubs, reflecting loss of epithelial identity and increased immune involvement.

Functional enrichment analysis elucidated the nature of this shift, highlighting a transition toward a chronic inflammatory state. Notably, the detailed mapping of the "Complement and coagulation cascades" pathway suggested that the tumor microenvironment actively promotes immune cell recruitment and adhesion while avoiding direct cell lysis, thereby creating conditions that sustain tumor progression.

Differential co-expression analysis further highlighted widespread regulatory rewiring, characterized mainly by loss of coordinated gene–gene interactions rather than the formation of new ones. Key genes such as *STRA6*, *PODXL*, and *CALB1* showed dramatic connectivity changes, suggesting altered control of metabolic and differentiation-related processes in KIRP.

At the patient level, Patient Similarity Networks revealed clear molecular subgroups within the tumor cohort. Integration of transcriptomic and mutational data through Similarity Network Fusion refined this stratification and identified communities associated with differences in survival, tumor mutational burden, and pathway alterations. Importantly, survival differences emerged despite not being used in the clustering process, supporting the clinical relevance of the identified subtypes.

Overall, our results show that KIRP heterogeneity is best captured by a systems-level perspective, where coordinated transcriptomic, genomic, and network alterations jointly define tumor behavior. This integrative framework provides a useful basis for molecular stratification and may support future studies aimed at prognosis, patient risk assessment, and the identification of biologically meaningful therapeutic targets.

Federico Lattanzio, Domenico Azzarito, Michele Pezza„

References

[1] Barabási, A.-L., and Oltvai, Z. N. Network biology: understanding the cell's functional organization. *Nature Reviews Genetics 5*, 2 (2004), 101–113.

[2] Berry, D. C., et al. The RBP4-STRA6 pathway is essential for maintaining colon cancer stem cells. *Cell Stem Cell 11*, 4 (2012), 549–560.

[3] Chen, Z., et al. GPX3 suppresses tumor progression and increases chemosensitivity in renal cell carcinoma. *Cancer Medicine 10*, 5 (2021), 1623–1636.

[4] De Coene, L., et al. Differential co-expression network analysis with DCoNA reveals isomir targeting aberrations in prostate cancer. *Bioinformatics 39*, 2 (2023), btad051.

[5] Hoadley, K. A., Yau, C., Hinoue, T., et al. Multiplatform analysis of 12 cancer types reveals molecular classification within and across tissues of origin. *Cell 158*, 4 (2014), 929–944.

[6] Hudson, N. J., Reverter, A., and Dalrymple, B. P. A differential wiring analysis of expression data correctly identifies the gene containing the causal mutation. *PLoS Computational Biology 5*, 5 (2009), e1000382.

[7] Kawaguchi, R., Yu, J., Honda, J., et al. A membrane receptor for retinol binding protein mediates cellular uptake of vitamin A. *Science 315*, 5813 (2007), 820–825.

[8] Krismann, M., et al. Calretinin as a marker for mesothelioma but not for renal cell carcinoma. *Histopathology 34*, 5 (1999), 433–439.

[9] Linehan, W. M., and Ricketts, C. J. The metabolic basis of kidney cancer. *Cancer Discovery 9*, 8 (2019), 1006–1021.

[10] Liu, B., et al. Podocalyxin promotes glioblastoma multiforme cell invasion and proliferation via β-catenin signaling. *PLoS ONE 15*, 1 (2020), e0227469.

[11] Taylor, I. W., Linding, R., Warde-Farley, D., et al. Dynamic modularity in protein interaction networks predicts breast cancer outcome. *Nature Biotechnology 27*, 2 (2009), 199–204.

[12] Vuong, L., et al. Tumor microenvironment dynamics in clear-cell renal cell carcinoma. *Cancer Discovery 9*, 10 (2019), 1349–1357.

[13] West, J., Bianconi, G., Severini, S., and Teschendorff, A. E. Differential network entropy reveals cancer system hallmarks. *Scientific Reports 2*, 1 (2012), 802.

[14] Wu, X., et al. STRA6 regulates tumor immune microenvironment and is a prognostic marker in papillary thyroid carcinoma. *Frontiers in Endocrinology 11* (2020), 543.

[15] Yu, H., Kim, P. M., Sprecher, E., Trifonov, V., and Gerstein, M. The importance of bottlenecks in protein networks: correlation with gene essentiality and expression dynamics. *PLoS Computational Biology 3*, 4 (2007), e59.